

Petrography of the quartz feldspar porphyry dyke from Patagundemgollapalle (Kadiri Mandal, Anantapur District), Andhra Pradesh, India

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ABSTRACT

Quartz Feldspar Porphyry (QFP) occurs as a dyke body intruding into the granitoid country, located near Namala Gudi temple, on the way to Kadiri from Pulivendula. It runs for about 6-8 km in the NNE-SSW direction. It has phenocrysts of quartz and feldspars in a fine grained matrix of the same material. Due to the higher incidence of K-feldspar, the rock has more pink color. At Namalagudi area, this shows extensive brittle deformation. All along the strike, the features remain same, but not the brittle deformation. The quartz appears to have two populations, viz., phenocrysts – the average size is 5 mm (maximum 7 mm) with the ground mass containing grains of less than 0.1mm. The larger grains appear to have nucleation for the growth of the crystal and commonly contain melt inclusions and secondary fluid inclusions.

Key words: QFP, Phenocrysts, K-feldspar, Brittle deformation, Fluid inclusions.

INTRODUCTION

The QFP (Quartz feldspar porphyry) (Rose, 2007) dyke body is located in the topographical map of 57J/3 of Survey of India. It is covered between the 14° 18' 48'' and 14° 20' 20'' North latitude and 78° 10' 21'' and 78° 12' 35'' East longitude, there by covering nearly an area of about 12 km². This appears to be the first report of QFP occurring as dyke body in this area. The nearest township is Pulivendula of the Kadapa district. This granitoid terrain is traversed by number of basic dykes ranging in composition from olivine dolerite to normal dolerite. But there is the conspicuous Quartz Feldspar Porphyry (QFP), seen intruding into the granitoid terrain and further going below the sediments of the Cuddapah Super group, indicating that it is younger than the granitoids and older than the Cuddapah sediments, i.e., the Gulcheru Quartzite of the Papaghni Group. The dyke body occurs as strike ridge dipping steeply towards the west. The outcrop occurring near the temple Namalagudi, is the major one. It is located on the right side of the Pulivendula–Kadiri new road (SH–28). This outcrop runs nearly for about 1.75km in the NNE - SSW direction and has a height of 20 m. After 1.75 km, it occurs as discontinuous outcrops and runs further for about 1km in the SSW direction having a height less than 20 m. The third QFP is located south of the Vepalapalle village, where the outcrops have an

almost east–west trend and runs for less than 500 m with a height of 20 m.

Petrologically, this rock consists of quartz and feldspar occurring as phenocrysts. The matrix is also of the same material but is of fine grained nature. Due to the higher incidence of K-feldspar, the rock has more pink color. On way to the village Subbanaguntapalle, the QFP has dark brown matrix with phenocrysts of feldspar. This is overlain by the pink variety of QFP. In addition, in the stream section, adjacent to the road on way to Subbanaguntapalle, there are a few outcrops of QFP with enclaves of green chloritic material. In addition, there are many dolerite dyke events that criss-cross the granitic terrain. In fact the basic dyke cuts the QFP, reflecting that the latter is the first event of dyking activity. The country rock is represented by peninsular gneiss in the southern part, intruded by younger various granitoid events in the northern part of the area under report. The event stratigraphy of the granitoids and the dyking activity has also been worked out. Quartz veins trending mainly east-west, represent the youngest phase of igneous activity.

GEOLOGICAL SETUP

Event stratigraphy of the granitoids (Early Units)

The country rock is mainly granitoid in nature. There

are nearly seven phases of granitic activity in this area, established mainly by the nature of occurrence in the form of xenoliths and cross cutting nature. The first two phases of granitoids, i.e., G1 and G2 (Figure 1a) are of mainly tonalitic in composition. The G3 is medium to coarse grained and is more like granodiorite with higher incidence of plagioclase feldspar (Figure 1b). The mafic mineral is mostly hornblende and its incidence increases to make it as mesocratic at places, which can be seen in good number in hills. The G4 is granitic in composition and can be called grey granite (Figure 1c). Compared to G3, this is fine grained and has limited occurrence. The G5 is fine grained pink granite that has intruded other granitoids (Figure 1d). Its end phase is pegmatoidal in nature and has impregnated the coarse grained pink feldspars in to the other granitoids. The G6 is more like leucogranite with insignificant mafic minerals (Figure 1e). It is fine to medium grained and occupies the maximum area under

the study. The entire western part of the study area is covered by this. The G7 is the pegmatoidal event (Figure 1f), related to G7. These may form the early units of the area (Javier Gil, 2010).

Dyking activity in the study area (late units)

All the dyke bodies when viewed in the satellite image, have similar spectral signatures, i.e., have similar tone and morphology (land form) – i.e., linear ridges, but slight difference in texture. Hence close and detailed field study is essential to record the differences between them. As mentioned before, the QFP forms one of the most interesting dyke body, hence forms the main aspect of the present study. The other dyke bodies are also studied. 5 dyking events have been identified by studying the cross cutting relation of the dykes. All the dyking events are considered as late units (Javier Gil, 2010).

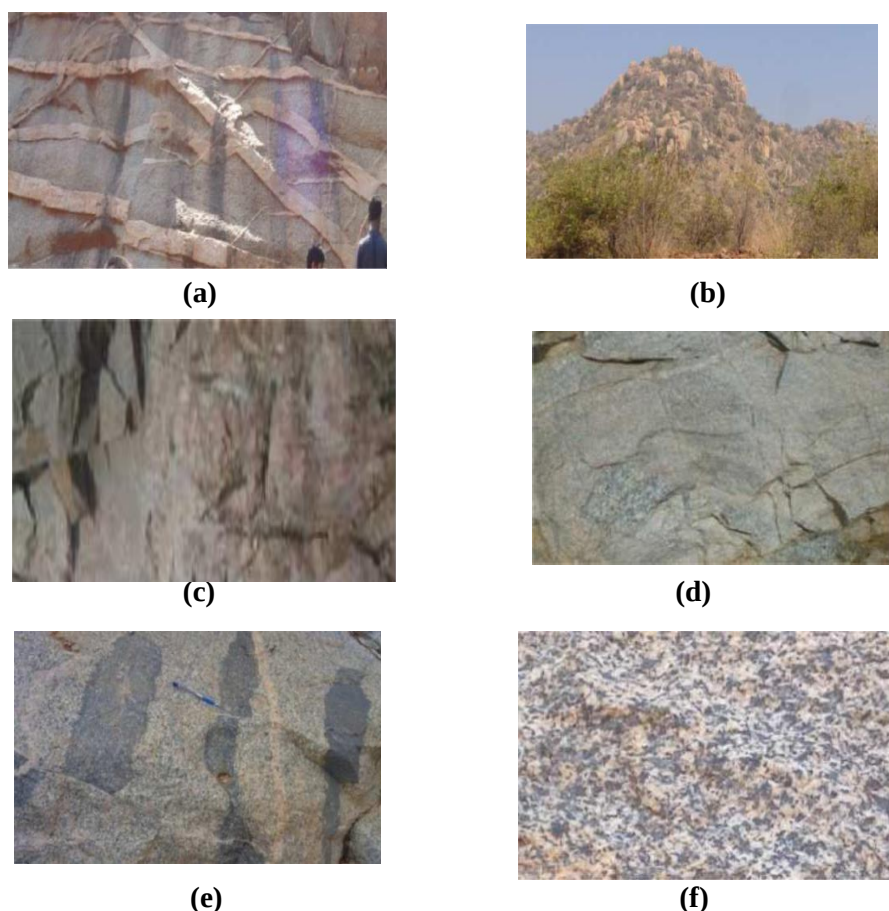


Figure 1. Event stratigraphy of the granitoids. (a) The youngest is the seventh phase G₇ is pegmatitic,

(b) The sixth phase G_6 is represented by leuco granite, (c) The fourth phase G_4 intrude by the fifth phase G_5 , (d) The third phase G_3 as enclave in the fourth phase G_4 , (e) The first phase, G_1 and G_2 in the third phase, G_3 . (f) The close-up of the G_3

Quartz Feldspar Porphyry Dyke (QFP)

Based on the field evidence, i.e., cross cutting relationship between the dykes, the relative ages of the dyking episodes have been worked out. The dykes of Quartz - Feldspar Porphyry (QFP) are considered to be of the first episode. (Dy1). Further, the 5 phases of dyking activity can be seen in the area through the Google Earth

satellite image (Figure 2).

The QFP has intruded into the country rock that is a granitoid. (Figure 3) The contact between the QFP and the country rock near the temple is marked by a brittle shear zone. It trends NNE-SSW and the rock is highly shattered. The contact also shows fine grained nature of QFP, reflecting the contact effect.

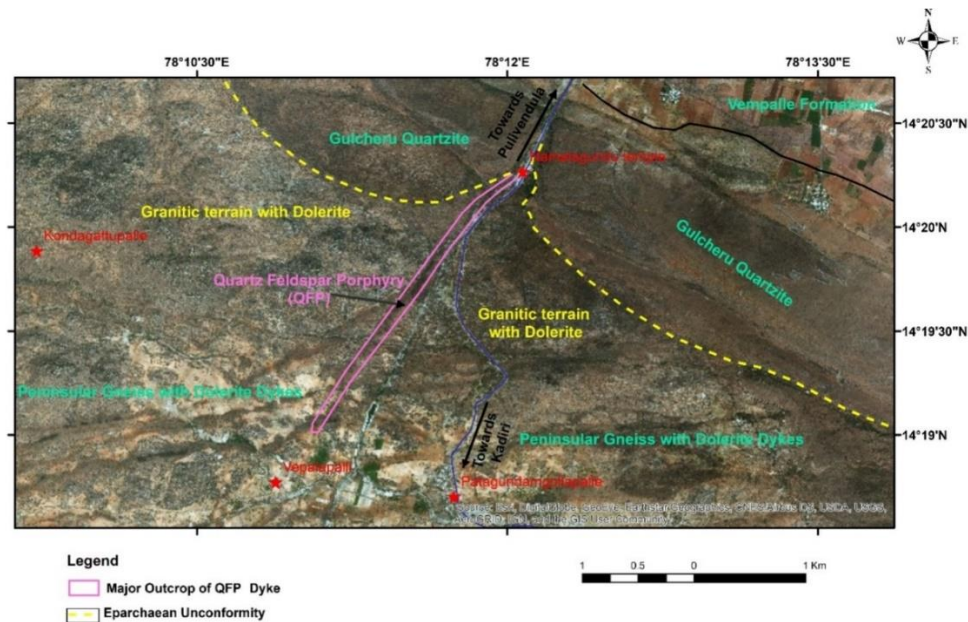


Figure 2. Study area showing QFP and other field details on Google Earth Image.



Figure 3. The field set up of the QFP (Dy1).

MEGASCOPIC CHARACTERISTICS

Near the temple Namalagudi area, the porphyry has flesh red color with the porphyries of K- feldspar, quartz and plagioclase feldspar (Figure 4). The matrix is fine to medium grained and is also pink in color, indicating the presence of fine to medium grained k-feldspar. The quartz appears to have two populations of phenocrysts (Figure 4) and the average size is 5 mm with the ground mass containing grains of less than 0.1mm. The larger grains measure up to 7mm and appear to have nucleation for the growth of the crystal. The quartz grains are sub-rounded (Leonard et al., 2005).

In the photograph (Figure 4) the quartz reflects half white color, whereas the K-feldspar reflects pink to



Figure 4. Megascopic features of QFP. (Field Feature).

MICROSCOPIC CHARACTERISTICS

Mineralogy

Amongst the major minerals quartz and feldspar form the major constituents. Anhedral quartz and anhedral to subhedral feldspar phenocrysts constitute major percentage in a thin section. Sub-rounded to rounded quartz is seen as phenocrysts, with a groundmass consisting of grains <0.1mm in size (Figure 6a-c). The minerals appear to have grown around a black colored nucleus. Similar views were expressed by Leonard et al., (2005).

Further, two populations of quartz is seen, (i) a large subhedral quartz phenocrysts in contact with smaller

light pink color. Matrix also varies from light pink to deep pink. The matrix at few places, that is seen at the lower of the hills of QFP, is fine grained and greenish material with phenocrysts and of quartz and feldspar, the later invariably show nuclei. There are phenocrysts that show sub-hedral to euhedral shape and the phenocrysts are mostly anhedral with angular fragments (Figure 5). There are a few feldspar crystals that show dark colored nuclei (Figure 5). This is seen on way to the village Subbanaguntapalle (Figure 5), where it is overlain by the pink variety of QFP (Figure 4). In addition in the stream section, adjacent to the road on way to Subbanaguntapalle, there are a few outcrops of QFP with enclaves of green chloritic material.



Figure 5. A variety of QFP with greenish grey matrix (Field Feature).

numerous quartz grains, and (ii) a large phenocryst with melt and fluid inclusions, where the surrounding melt was seen trapped by the growing quartz along the grain boundary (Leonard et al., 2005). These features are reflected in the Photomicrographs (Figures 7a-b). Feldspars are present in major proportions in every sample, but have undergone extensive sericitization. Orthoclase phenocrysts are frequently glomeroporphyritic and form clusters of numerous crystals (Figure 8). Inclusions of aggregates of opaques and titanite (?) are found in the cores of many orthoclase crystals (Leonard et al., 2005).

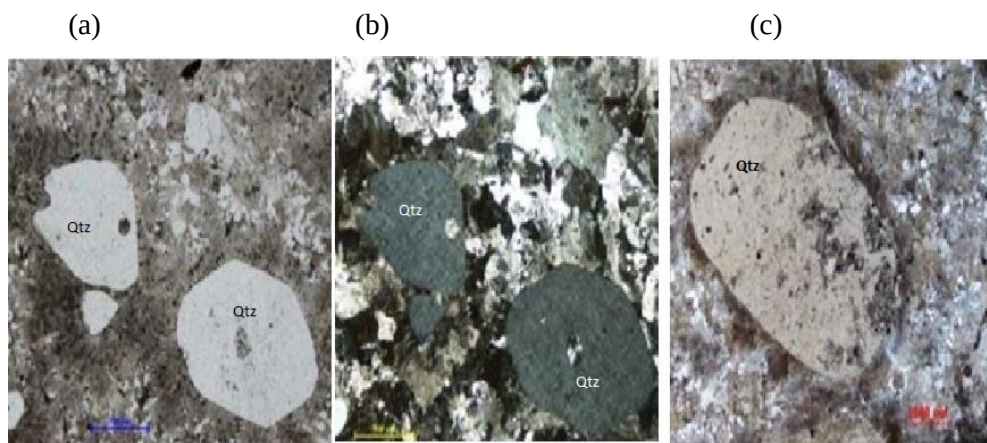


Figure 6. Photomicrographs showing euhedral and subhedral quartz (Qtz), with secondary fluid inclusions. The subhedral quartz reflects that they are corroded and sorbed (Figures a,b) (a) Under polarized light, (b) under crossed nicols, (c) Under crossed nicols.

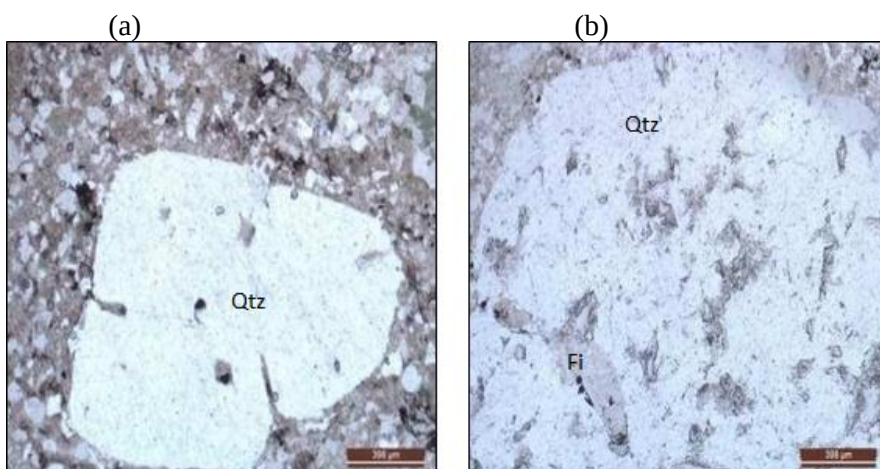


Figure 7. (a) Photomicrographs reflecting large subhedral quartz Phenocryst in contact with smaller numerous quartz grains, (b) Photomicrographs showing a large phenocryst with melt and fluid inclusions (Fi).

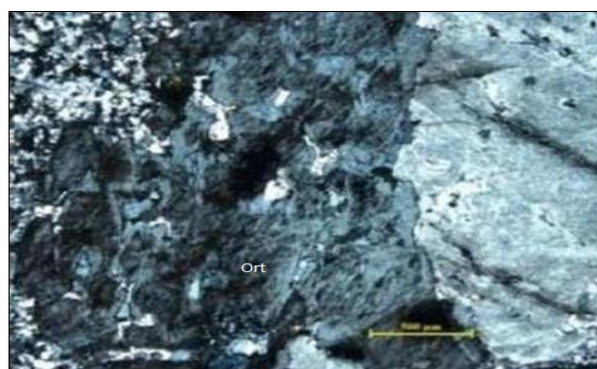


Figure 8. Photomicrograph reflecting the Glomeroporphyritic orthoclase (Ort) (form of clusters of numerous crystals) - Under crossed nicols.

Groundmass / Matrix

The ground mass is constituted mainly by the quartz and feldspar of minor size and are mostly subhedral to

euhedral, as observed in the figures (6) to (9). Inclusions of apatite, zircon, chlorite, Fe- sulphide and rutile are observed (Figure 9a-9f). The dyke's groundmass mineralogy generally consists of fine

grained quartz, plagioclase and orthoclase with accessory biotite, chlorite, muscovite, pyrite, and calcite, and rutile, ilmenite. Chlorite appears to be local alteration product of reddish brown biotite. Chlorite is

associated with calcite and arrow calcite veins occupy micro fractures, suggesting late paragenesis. Individual calcite grains contain pyrite inclusions (Leonard et al., 2005).

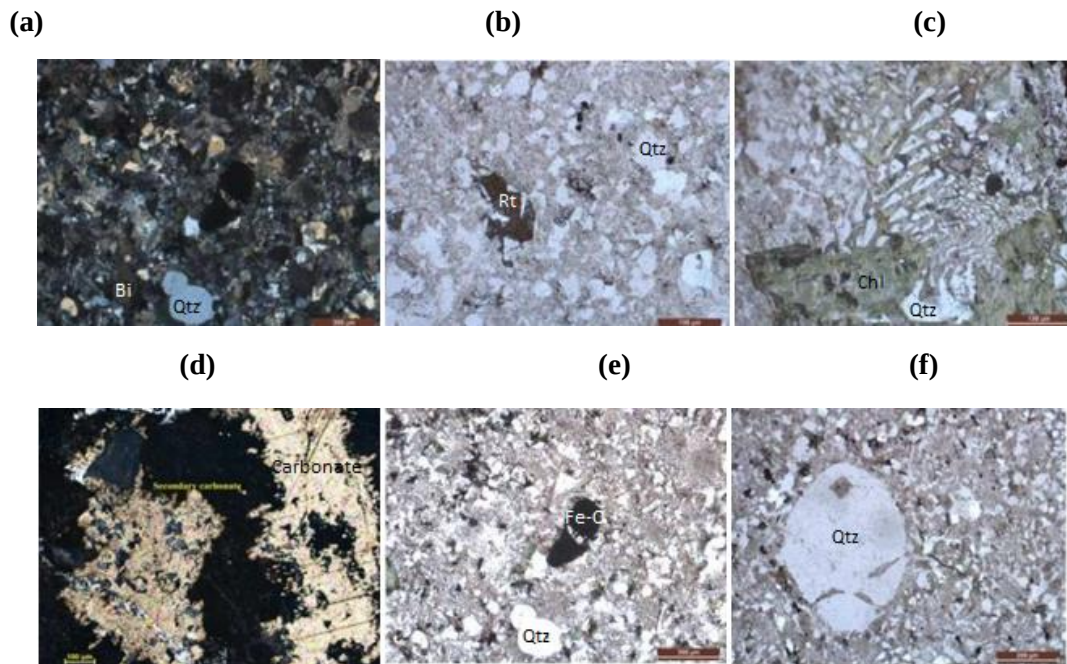


Figure 9. Photomicrographs of Quartz Feldspar Porphyry. (a) Biotite (Bi) with other accessories, (b) Rutile (Rt) with other accessories, (c) Chlorite (Chl) with other accessories, (d) Carbonate with other accessories, (e) Fe-oxide (Fe-O) with other accessories, (f) Quartz (Qtz) with melt-inclusions

Texture

The study of textures is essential to understand the history of cooling and solidification (Bryon et al 1995). The Quartz Feldspar Porphyry exhibits resorbed quartz phenocrysts, graphic texture and granophyric textures (Streckinson, 1974; Vernon, 2018). Resorbed texture is very well reflected by the quartz phenocrysts (Figure

10). The margins of the clasts / crystals are affected by the residual liquid, resulting in the formation of dents in the outline of the mineral. Graphic or runic intergrowth between quartz and feldspar (Leonard et al., 2005) is very conspicuous in almost all the thin sections (Figure 11). Granophyric intergrowth between quartz and feldspar is also seen in almost all the thin sections (Figure 12).

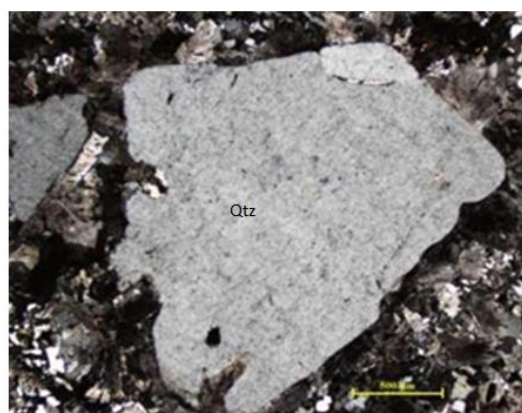


Figure 10. Photomicrographs showing resorbed quartz phenocryst.

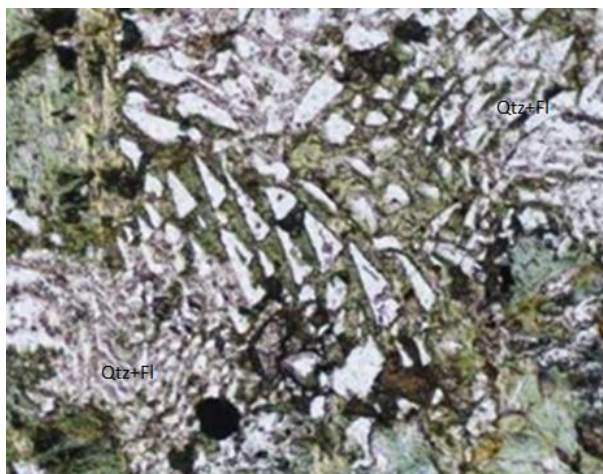


Figure 11. Photomicrograph showing graphic or runic intergrowth between quartz (Qtz) and feldspar (Fl).

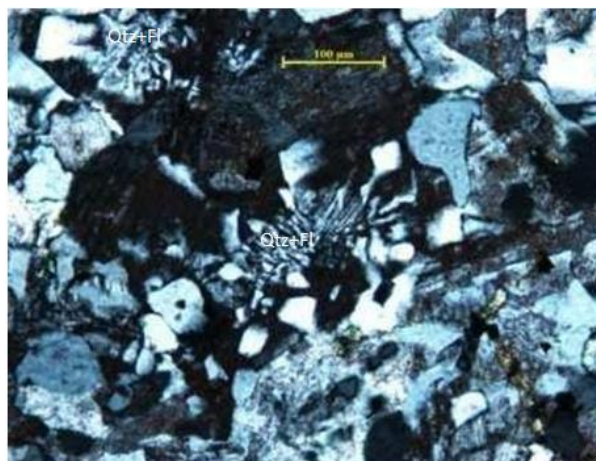


Figure 12. Photomicrograph showing granophyric intergrowth between quartz (Qtz) and feldspar (Fl).

CONCLUSIONS

The study of the QFP has led to the following conclusions.

- (i) It is a unique rock in the local geological set up.
- (ii) There are seven events of the granitoid activity that can be recognized as early units.
- (iii) Intruding the granitoids is the QFP, that has very clear reflection of porphyritic nature and it is seen as a first phase of intrusive into the granitoid country. In addition to this, there are nearly four phases of basic intrusives that could be identified by the crosscutting relation in the field. These are recognized as late units.
- (iv) The phenocrysts are K-feldspar, quartz and

plagioclase feldspar. Quartz has two populations, viz., the smaller and the bigger. The main outcrop is pink in color with the matrix reflecting light pink to pink color indicating the same composition, (higher incidence of k-feldspar) but finer in grain size.

(v) Another variety of QFP, with dark grey matrix and phenocrysts of plagioclase feldspar is noticed occurring below the pink variety. The phenocrysts have dark colored nucleus.

(vi) Fluid inclusions are seen in almost all phenocrysts. Texturally, this rock exhibits glomeroporphyritic K-feldspar and graphic and granophyric intergrowths between quartz and alkali feldspar are also seen extensively.

ACKNOWLEDGEMENTS

The authors express their sincere thanks to the reviewers for their valuable comments towards improving the quality of the manuscript. We are also grateful to the Chief Editor of JIGU for useful suggestions and appropriate editing.

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Received on: 8.3.19; Revised on: 17.5.19; Accepted on: 26.5.19

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